

# Staines Rajith

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**Address:** London, United Kingdom

## EDUCATION

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**Robotics and Artificial Intelligence MSc**  
**Queen Mary University of London**

**Sep 2025 – Sep 2026**

- **Key Modules:** Cognitive Robotics, Machine Learning and Artificial Intelligence, Medical Robotics and Surgical Techniques, Deep Learning for Data and Image Analysis
- **Key Projects:** Electro-optical PPG system, Neural Network model predicting pedestrian behaviour, Few-Shot Learning in Medical Imaging

**Robotics, Mechatronics and Control Engineering BEng (Hons)**  
**Loughborough University**

**Oct 2021 – Jun 2024**

- **Key Modules:** State Space Control, Manufacturing Automation and Control Systems, Electronics, Programming, Interfacing with Mechatronics Systems.
- **Key Projects:** Eco-Friendly Intra-Campus Monorail System Design, Gymnastics Stamp-bug and Rover, Vision-Based Robotic Arm Control, Roll-angle Measurement Device

## TECHNICAL SKILLS

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**Electrical & Control:** Siemens PLC, CODESYS, Structured Text, Ladder Logic, EasyEDA, NI Multisim, Microcontrollers, Sensors and Actuators

**Programming:** Arduino, HTML, Python, C++, ROS, MATLAB.

**CAD:** Fusion 360, Siemens NX, AutoCAD 2D, CATIA, Engineering Drawings.

**Software:** Simulink, Gazebo.

## WORK EXPERIENCE

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**First Quantum Mineral Ltd – Control Engineering Intern**  
**Jul 2022 – Aug 2022**

- Developed and implemented PLC control logic using structured text and ladder logic in CODESYS, defining I/O behaviour and gaining practical experience in system control, automation and safe-state design.
- Designed and tested a functional traffic light system, applying structured systems engineering methods and fault testing to meet requirements and deliver a working, repeatable prototype.

**Trident College – Teaching Assistant**  
**Jan 2021 – Mar 2021**

- Supported students in lab-based learning, reinforcing engineering concepts and ensuring safe, structured experimentation.
- Delivered tutoring and assisted teachers with grading, improving student engagement and developing clear technical communication skills.

**First Quantum Mineral Ltd – Technical Team Assistant**  
**Nov 2018 – Dec 2018**

- Monitored and maintained machinery in copper smelting operations, optimising air cooler functionality and airflow efficiency to improve furnace performance and sustain 98% equipment uptime.
- Identified and reported equipment abnormalities to supervisors, preventing operational disruptions through rigorous adherence to safety and maintenance protocols.

## PROJECTS

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### Photoelastic Tactile Sensor for a Robotic Palm (Collaboration with Shadow Robotics)

- Designed and built a standalone palm prototype combining a photoelastic membrane with a folded optical system to address robotic palm space constraints, then developed automated indentation and synchronised data collection for neural-network calibration.
- Evaluated four machine learning architectures for force estimation and contact localisation, achieving a lowest normal-force MAE of 0.244 N and sub-millimetre localisation with separate models on held-out contact locations, demonstrating the prototype's tactile sensing capability under controlled normal indentation.

### Mars Rover for UKSEDS Olympus Rover Trials

- Led the electrical subsystem as Vice Lead and Electrical Team Lead for a student-built Mars rover, coordinating hardware integration that contributed to the team placing 3rd overall in the UKSEDS Olympus Rover Trials Finals at RAL Space.
- Designed power distribution and integrated Raspberry Pi 3 control, LoRa communications, dual 3S LiPo power management, IR obstacle detection, and a camera vision system into a robust, serviceable mobile robotic platform.
- Conducted system-level integration, fault testing and validation, identifying and resolving hardware and communication issues to improve commissioning readiness and reliability during field testing and competition preparation, while contributing to the team receiving the Best Critical Design Review (CDR) Award.

### Vision-Based Robotic Arm Control

- Engineered a physical robotic arm system for precision pick-and-place operations of coloured building blocks, using ROS on Linux to coordinate camera feedback and kinematic execution.
- Implemented machine learning algorithms for the integrated vision system, training the network on an extensive dataset of colours and shades to ensure robust object recognition under varying lighting conditions.
- Evaluated real-world system performance with Python by analysing kinematic positioning accuracy and visual classification metrics, validating both system reliability and environmental adaptability.

## CERTIFICATIONS

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|-----------------------------------------------------------------|----------------------------------------------------------------|
| • <b>EV Engineering Advanced Job Simulation</b> – Forage (Ford) | • <b>Career Essentials in Software Development</b> – Microsoft |
| • <b>Machine Learning at the edge</b> – Arm                     | • <b>AutoCAD 2D</b> – CADD Centre Training Services Pvt. Ltd   |
| • <b>C++ programming</b> – OpenEDG C++ Institute                | • <b>Python</b> – Croma Campus Pvt. Ltd                        |
| • <b>Programming with Python</b> – OpenEDG Python Institute     | • <b>AWS Release and Deploy</b> – edX                          |
|                                                                 | • <b>PLC Programming</b> – Udemy                               |

## EXTRACURRICULARS

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### Volunteering

- Coordinated annual day trips from Trident College to Maheba Refugee Camp (UNHCR) (2018–2019).

### Activities

- Cycling, Cross Country, Squash, Rugby, Cricket, Music.